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1. PURPOSE

This document provides the steps to create virtual Optical drive in VIO server. VOD is act as DVD drive, help to install, upgrade OS in AIX environment where NIM operation was unable to perform.

2. SCOPE

2.1. In scope:

2.1.1 We will receive RITM to create Virtual Optical drive and record all activity in RITM.

3. PREREQUISITES

3.1 Operating System AIX BOS ISO Images. We can download it from URL https://www.ibm.com/servers/eserver/ess/ProtectedServlet.wss#toggle

3.2 VIO server should have enough free space in rootvg to create virtual media repository.

4. PROCEDURE

4.1 Login to VIO server with MID and switch to padmin with password.

4.2 Check whether existing media repository exists or not by executing below command.

$ lsrep

```
$ lsrep
The DVD repository has not been created yet.
```

4.3 If repository not exists, create a repository by issuing below command

$ mkrep -sp rootvg -size 28G

```
$ mkrep -sp rootvg -size 28G
$
```

Note: By default, above command will create a Logical Volume “VMLibrary_LV” in the volume group “rootvg” with size of 28GB. Also it will create a mount point “/var/vio/VMLibrary”

4.4 Check the VMLibrary repository by issuing below command.

$ df -gt /var/vio/VMLibrary

```
$ df -gt /var/vio/VMLibrary
Filesystem    GB blocks      Used      Free %Used Mounted on
/dev/VMLibrary     28.00     22.64      5.36   81% /var/vio/VMLibrary
```

4.5 Switch to root user and Mount the NFS share (where the BOS iso image persists) in the

mount point "/mnt" of VIO server. We have ISO images in NIM server filesystem "/mksysb" which was exported.

$ r_oem

mount -v nfs u0bnim001:/mksysb /mnt

```
# mount -v nfs u0bnim001:/mksysb /mnt
# df -gt /mnt
Filesystem    GB blocks      Used      Free %Used Mounted on
u0bnim001:/mksysb   3800.50   1766.60   2033.90   47% /mnt
```

4.6 Exit from root, then Create Virtual optical media file in the Virtual Media Repository with BOS iso images by issuing below command.

exit
***$ mkvopt -name AIX72_VOL1 ***
-file /mnt/Aix7200-02-04-1914_Install_DVD_1_of_2_82019_LCD8257801.iso -ro
***$ mkvopt -name AIX72_VOL2 ***
-file /mnt/Aix7200-02-04-1914_Install_DVD_2_of_2_82019_LCD8257901.iso -ro

```
$ mkvopt -name AIX72_VOL1 -file /mnt/Aix7200-02-04-1914_Install_DVD_1_of_2_82019_LCD8257801.iso -ro
$
```

```
$ mkvopt -name AIX72_VOL2 -file /mnt/Aix7200-02-04-1914_Install_DVD_2_of_2_82019_LCD8257901.iso -ro
$
```

Note: /mnt/Aix7200-02-04-1914_Install_DVD_1_of_2_82019_LCD8257801.iso and /mnt/Aix7200-02-04-1914_Install_DVD_2_of_2_82019_LCD8257901.iso are BOS iso images. AIX72_VOL1 and AIX72_VOL2 are Virtual optical media file

4.7 Create File based Optical device by execute the below command.

$ mkvdev -fbo -vadapter vhost2

```
$ mkvdev -fbo -vadapter vhost2
$
```

Note: This command creates virtual optical target device called vtopt0(FBO target => file-backed optical device) and map it to in the vhost2 adapter.

4.8 Load the Virtual media file to Virtual target device by issuing below command.

$ loadopt -vtd vtopt0 -disk AIX72_VOL1

```
$ loadopt -vtd vtopt0 -disk AIX72_VOL1
$
```

4.9 Now the Virtual repository is connected to the client machine/LPAR. To check the mapping by executing below command

$ lsmap -vadapter vhost2

```
$ lsmap -vadapter vhost2
SVSA            Physloc                                         Client Partition ID
--------------- ----------------------------------------------- ------------------
vhost2          U9117.MMB.101435R-V1-C204                       0x00000003

VTD                   0bnim02X0E03T
Status                Available
LUN                   0x8200000000000000
Backing device        hdisk7
Physloc               U78C0.001.DBJM826-P2-C1-T1-W5001738075660150-L8000000000000
Mirrored              false

VTD                   vtopt0
Status                Available
LUN                   0x8100000000000000
Backing device        /var/vio/VMLibrary/AIX72_VOL1
Physloc
Mirrored              N/A
```

4.10 To unload the Virtual media file from Virtual target device by issuing below command.

$ unloadopt -vtd vtopt0

```
$ unloadopt -vtd vtopt0
$
```

4.11 Then load the another Virtual media file to the virtual optical device, by issuing the below command.

$ loadopt -vtd vtopt0 -disk AIX72_VOL2

```
$ loadopt -vtd vtopt0 -disk AIX72_VOL2
$
```

5. VERSION HISTORY

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